# Loft

**User guide** · IngeTrazo plugin · `loft.py` · Ctrl+Alt+L

Makes a closed, printable solid that blends between two or more flat shapes, such as a square at the bottom and a circle above it.

## Installing and updating

1. Open File Explorer, paste `C:\Users\eddy_\AppData\Roaming\ingetrazo\plugins` into the address bar and press Enter.
2. Copy `loft.py` into that folder. When updating, choose **Replace the file**.
3. Restart IngeTrazo. Settings are remembered only while IngeTrazo is open, so a restart also clears old settings.

Keep only one copy of the plugin in the folder. Two copies under different names may both load.

**Shortcut:** Ctrl+Alt+L

**From the Python console:** paste the file's code in, select what the plugin needs, then type `run()`. If it can't find the 3D view by itself, use `run(viewport)`.

## Using it

1. Draw the shapes, for example a square with a circle above it.
2. Select their faces and press **Ctrl+Alt+L**.
3. Check the settings and click OK. The solid goes in a new group; your original shapes are left in place to delete or hide.

The shapes must face each other, one above the other or one in front of the other. Two shapes lying side by side on the ground give an error. Only the outer edge of each shape is used; holes are ignored.

## Settings

| Setting | Effect |
|---|---|
| Points per profile | Points around each shape. Defaults to at least 64 for curved lofts; it can't go below the shape with the most corners |
| Sections between profiles | Bands up the solid: 1 for Straight, 32 for S-curve, 24 for Smooth by default |
| Path | **Smooth S-curve** (sides leave and meet each shape square-on), **Straight**, or **Smooth curve through all profiles** for 3 or more shapes |
| Line profiles up automatically | Matches corners to the nearest part of the next shape, so the solid doesn't twist |
| Extra twist | Turns the last shape by this many points, to add a twist on purpose |

| Setting | Effect |
|---|---|
| Cap the first / last profile | Closes the ends; both must be ticked for a printable solid |
| Side faces | Auto splits only clearly bent side faces into triangles |
| Smooth finish | Hides the edge lines between faces that meet at less than the set angle (30° by default) |
| Group name | Name of the new group |

Changing **Path** fills in that path's suggested numbers; you can type your own afterwards.

## Smoothness

IngeTrazo shades each face flat, so a smooth surface can still look faintly faceted on screen. For a smoother look, raise the points to about 96 and the sections to about 48, about 9,000 faces. The printed part follows the true shape either way.

## Troubleshooting

| Problem | Cause | Fix |
|---|---|---|
| "Select two or more profile faces" | Fewer than two faces selected | Select both shape faces |
| "Profile lies side-on" | Shapes don't face each other | Place one shape above, or in front of, the other |
| "The profiles are in the same place" | Shapes overlap | Move one shape away, for example straight up |
| The solid twists | Automatic line-up is off, or extra twist is set | Tick **Line profiles up automatically** and set Extra twist to 0 |
| "Could not find the mesh's edges" in the console | IngeTrazo didn't expose the edges | The solid is fine; only the smooth finish wasn't applied |